Calculating Positive Predictive Value (PPV)

Derivation from Sensitivity, Specificity, and Prevalence

Medical AI & Clinical Statistics Technical Brief

August 24, 2026

1. Core Definitions

Let pos denote an **actually positive** ground-truth case, neg denote an **actually negative** case, and $\widehat{pos}$ denote a **predicted positive** model output.

Diagnostic Metric Formulations

- **Sensitivity (True Positive Rate):**

$$\text{Sensitivity} = P(\widehat{pos} \mid pos)$$

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- **Prevalence (Prior Probability):**

$$\text{Prevalence} = P(pos) \implies P(neg) = 1 - \text{Prevalence}$$

2. Deriving PPV Step-by-Step (Bayes' Rule)

Target Definition: $\text{PPV} = P(pos \mid \widehat{pos})$

Step 1: Apply Bayes' Rule

$$\text{PPV} = \frac{P(\widehat{pos} \mid pos) \times P(pos)}{P(\widehat{pos})}$$

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Step 2: Expand the Denominator via Total Probability

The total predicted positives $P(\widehat{pos})$ is the sum of True Positives (TP) and False Positives (FP):

$$P(\widehat{pos}) = P(\text{True Positives}) + P(\text{False Positives})$$

$$\text{PPV} = \frac{P(\widehat{pos} \mid pos) \times P(pos)}{P(\text{True Positives}) + P(\text{False Positives})}$$

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Step 3: Express Components in Terms of Diagnostic Metrics

- **Numerator / True Positives:**
  $P(\text{True Positives}) = P(\widehat{pos} \mid pos) \times P(pos) = \text{Sensitivity} \times \text{Prevalence}$
- **False Positives:**
  $P(\text{False Positives}) = P(\widehat{pos} \mid neg) \times P(neg) = (1 - \text{Specificity}) \times (1 - \text{Prevalence})$

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Final Substituted Equation

$$\text{PPV} = \frac{\text{Sensitivity} \times \text{Prevalence}}{(\text{Sensitivity} \times \text{Prevalence}) + (1 - \text{Specificity}) \times (1 - \text{Prevalence})}$$

3. Numerical Example: Breast Cancer Screening

Scenario Constraints:

- **Sensitivity** $= 90\%$ (0.90)
- **Specificity** $= 95\%$ $(0.95) \implies 1 - \text{Specificity} = 0.05$
- **Prevalence** $= 1\%$ $(0.01) \implies 1 - \text{Prevalence} = 0.99$

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Step 1: Calculate Numerator (True Positives)

$$\text{Numerator} = 0.90 \times 0.01 = 0.0090$$

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Step 2: Calculate False Positives

$$\text{False Positives} = (1 - 0.95) \times (1 - 0.01) = 0.05 \times 0.99 = 0.0495$$

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Step 3: Calculate Final PPV

$$\text{PPV} = \frac{0.0090}{0.0090 + 0.0495} = \frac{0.0090}{0.0585} \approx \mathbf{15.38\%}$$

Takeaway: Even with high specificity (95%), low prevalence (1%) causes false positives to outweigh true

4. Population Grid Breakdown (N = 10,000 Patient Cohort)

	Actual Breast Cancer (+)	**Healthy / Normal (-)**	**Total**
Test Positive ($\widehat{pos}$)			

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	Actual Breast Cancer (+)	**Healthy / Normal (-)**	**Total**
Test Positive ($\widehat{pos}$)	**90** (TP)	**495** (FP)	

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	Actual Breast Cancer (+)	**Healthy / Normal (-)**	**Total**
Test Positive ($\widehat{pos}$)	**90** (TP)	**495** (FP)	**585**
Test Negative ($\widehat{neg}$)			

4. Population Grid Breakdown (N = 10,000 Patient Cohort)

	Actual Breast Cancer (+)	**Healthy / Normal (-)**	**Total**
Test Positive ($\widehat{pos}$)	**90** (TP)	**495** (FP)	**585**
Test Negative ($\widehat{neg}$)	10 (FN)	9,405 (TN)	9,415
Total Population	**100**	**9,900**	**10,000**

Table: Structured 2x2 Matrix showing why PPV $= \frac{90}{585} = 15.38\%$.